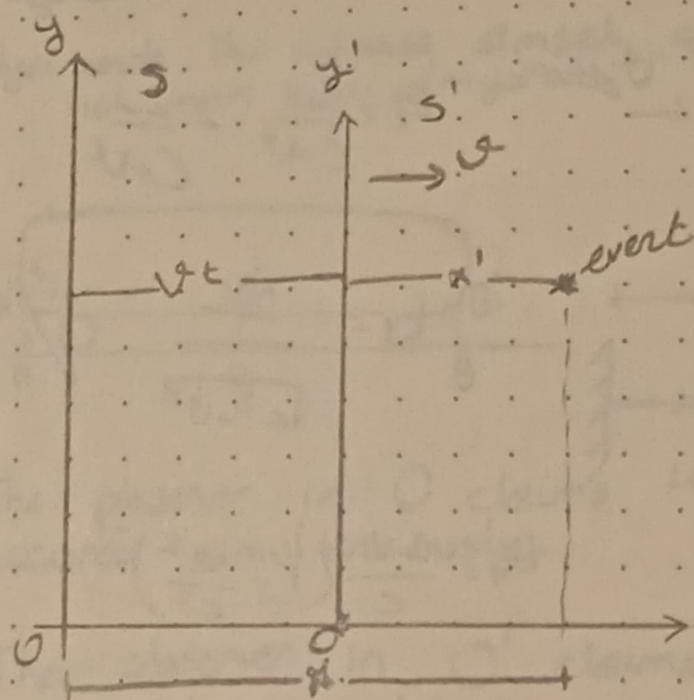




consider two reference frames

$$S(x, y, z, t)$$

$$S'(x', y', z', t')$$

$$\frac{dx'}{dt} = \frac{dx}{dt} - v$$

$$x' = x - vt$$

$$y' = y$$

$$z' = z$$

$$t' = t$$

$$x' = x - vt \quad (\text{Galilean Transformations})$$

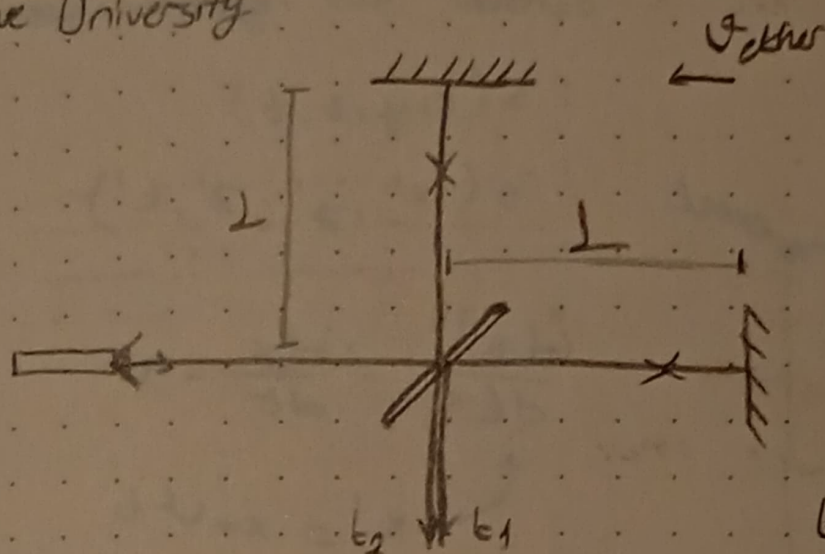
Newtonian Axioms

- There exists an absolute space in which Newton principles are true.
- All inertial frames share a universal frame

Galilean Relativity

- The laws of mechanics must be the same in all inertial frames of reference.

Ether: A medium which is itself devoid of all mechanical and kinematical qualities, but helps to determine mechanical (and electromagnetic) events.



$$t_1 = \frac{L}{c-v} + \frac{L}{c+v}$$

$$t_2 = \frac{2L}{\sqrt{c^2 - v^2}}$$

$$t_1 = \frac{2L}{c} \left(1 - \frac{v^2}{c^2}\right)^{-1}$$

$$t_2 \approx \frac{2L}{c} \left(1 + \frac{v^2}{2c^2}\right)$$

$$t_1 \approx \frac{2L}{c} \left(1 + \frac{v^2}{c^2}\right)$$

$$\Delta t = t_1 - t_2 \approx \frac{2L}{c} \left(\frac{v^2}{c^2} - \frac{v^2}{2c^2} \right)$$

Now, suppose the apparatus is rotated 90° so that arm 1 is now transverse to the ether wind. The new denote is t_1' and t_2' :

$$t_1' \approx \frac{2L}{c} \left(1 + \frac{v^2}{2c^2}\right) \text{ and } t_2' \approx \frac{2L}{c} \left(1 + \frac{v^2}{c^2}\right)$$

$$\Delta t' = t_1' - t_2' = -\frac{Lv^2}{c^3}$$

This is the time change produced by rotating the apparatus

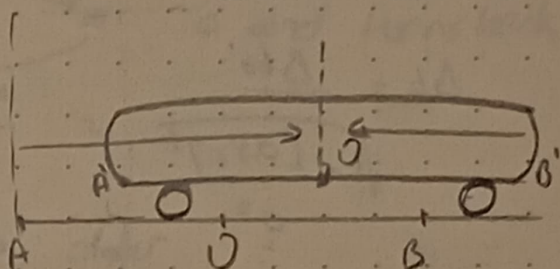
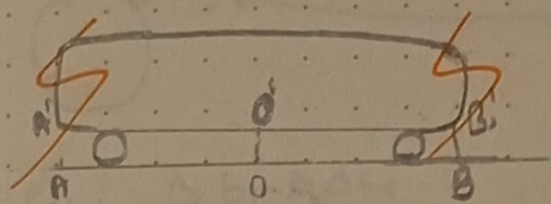
$$\Delta t - \Delta t' = \frac{2Lv^2}{2c^3} - \left(-\frac{Lv^2}{c^3}\right) = \frac{Lv^2}{c^3}$$

$$\Delta t = \frac{Lv^2}{c^3}$$

Time interval between t_1 and t_2
(Not found)

Einstein's Principle of Relativity

Aynı anda iki noktada simsek çakıyor.
ışık herin hızını alıyor.



The observer in O claims that A-B lightnings occurred simultaneously.

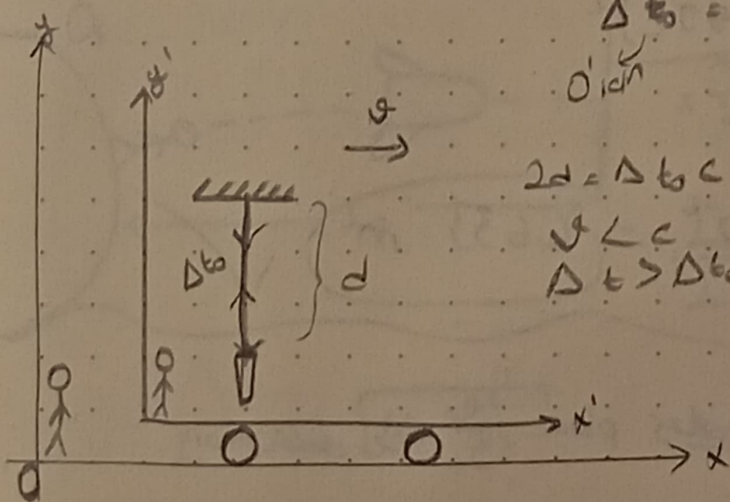
The observer in O' claims that lightning B occurred before

— The laws of physics must be the same in all inertial reference frames.

— The speed of light in vacuum has the same constant value c in all inertial reference frames regardless of the velocity of the observer.

Consequences of Einstein Relativity

• Time Dilation

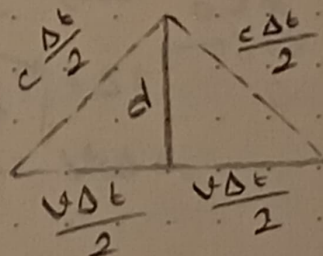


$$\Delta t_0 = \frac{2d}{c}$$

$$2d = \Delta t_0 c$$

$$v < c$$

$$\Delta t > \Delta t_0$$



$$d^2 + \left[\frac{v \Delta t}{2} \right]^2 = \left[\frac{c \Delta t}{2} \right]^2$$

$$\Delta t = \frac{2d}{\sqrt{c^2 - v^2}}$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

A spacecraft is moving towards earth with $v = 0.8c$.
A clock is ticking per sec. How many seconds is the clock behind the clocks on earth per min?

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} = \frac{5}{3} \Delta t_0$$

$$[1 \text{ sec}]^0 = [1.6 \text{ sec}]^{0'}$$

0.6 sec behind

36 secs behind per min

Muons are unstable particles with lifetime $\tau_\mu = 2.2 \times 10^{-6} \text{ s}$.
Can they reach earth surface if they're created in the atmosphere?

$$x = vt \quad \lambda = (0.99c) \times (2.2 \times 10^{-6})$$

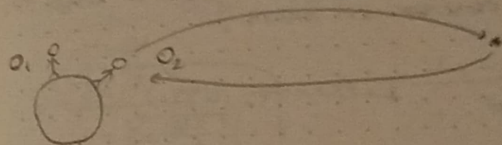
$$= 653 \text{ m} \rightarrow \text{they can't in Newton's physics}$$

$$t = \gamma t_0 \rightarrow \frac{2.2 \times 10^{-6}}{\sqrt{1 - \left(\frac{0.99c}{c}\right)^2}} = 15.6 \times 10^{-6} \text{ s}$$

$$x' = 0.99 \times 15.6 \times 10^{-6} = 4633 \text{ m} \quad \checkmark$$

Length Contraction only takes place in the direction of motion.

Twin Paradox



$O_1: \Delta t_2 > \Delta t_1$ O_1 gets older

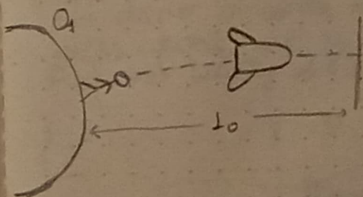
$O_2: \Delta t_1 > \Delta t_2$ O_2 gets getting older

What actually happens is that O_1 gets older and the time for O_2 will be dilated.

Resolution: O_2 is not experiencing an inertial frame of reference. Therefore, special theory of relativity does not apply.

Length Contraction

The measured distance between two points in the space also depends on the frame of reference of the observer



$$O_1: L_0 = v \cdot \Delta t \rightarrow \gamma t_0$$

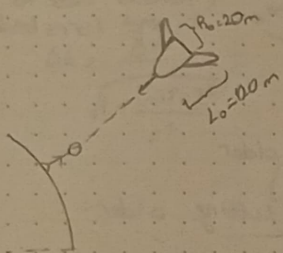
$$O_2: L = v \cdot \Delta t_0$$

$$\frac{L_0}{L} = \frac{\gamma t_0}{\Delta t_0} \rightarrow L = \frac{L_0}{\gamma}$$

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$L \leq L_0$$

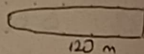
What is the length of a spacecraft whose rest length $L_0 = 120$ m and 20 m wide at $v = 0.99c$ relative to an observer on earth?



$$L = 120 \text{ m} \sqrt{1 - (0.99c)^2/c^2}$$

$$L = 17 \text{ m}$$

$$R = 20 \text{ m}$$



Lorentz Transformation Equations

The space-time transformations valid from $v=0$ to $v \approx c$ can be calculated by LT equations

$$\begin{array}{l|l} S \rightarrow S' & S' \rightarrow S \\ x' = \gamma(x - vt) & x = \gamma(x' + vt') \\ y' = y & y = y' \\ z' = z & z = z' \\ t' = \gamma(t - \frac{v}{c^2}x) & t = \gamma(t' + \frac{v}{c^2}x') \end{array}$$

Lorentz Velocity TE

Again we assume that S' is the frame moving at a constant speed v relative to S .

$$\frac{dx'}{dt'} = \frac{\gamma(dx - vdt)}{\gamma(dt - \frac{v}{c^2}dx)} \rightarrow \frac{dx'}{dt'} = u_x' = \frac{dx - vdt}{dt - \frac{v}{c^2}dx}$$

$$= \frac{\left(\frac{dx}{dt}\right) - v}{1 - \frac{v}{c^2}\left(\frac{dx}{dt}\right)} \rightarrow u_x' = \frac{u_x - v}{1 - \frac{u_x v}{c^2}}$$

$$u_x' = \frac{u_x - v}{1 - \frac{u_x v}{c^2}}, \quad u_y' = \frac{u_y}{\gamma(1 - \frac{u_x v}{c^2})}, \quad u_z' = \frac{u_z}{\gamma(1 - \frac{u_x v}{c^2})}$$

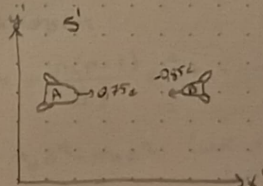
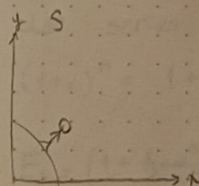
• Again $v \ll c$, we obtain the Galilean velocity transformation equations.

• If you would like to reverse the frames,

$$u_x = \frac{u_x' + v}{1 + \frac{u_x' v}{c^2}}$$

Example

An observer on earth measures two spacecraft velocities as $u_A = 0.75c$ and $u_B = -0.85c$ on the same direction. Find the velocity of B relative to the observer in A.



$$u = \frac{u' - v}{1 - \frac{u' v}{c^2}} = 0.977c$$

$$\frac{-0.85c - 0.75c}{1 - \frac{(-0.85)(0.75)c^2}{c^2}}$$

The relativistic kinetic energy can be obtained by

$$W = E = \int \vec{F} \cdot d\vec{s} = \int \frac{d(\gamma m_0 u)}{dt} ds = \int \left(\frac{ds}{dt} \right) d(\gamma m_0 u)$$

$$E = \int u \, d(\gamma m_0 u) \quad E = \frac{m_0 u^2}{\sqrt{1 - \frac{u^2}{c^2}}} - m_0 \int \frac{u \, du}{1 - \frac{u^2}{c^2}}$$

$$\int x \, dy = xy - \int y \, dx$$

$$E = \frac{m_0 u^2}{\sqrt{1 - \frac{u^2}{c^2}}} + \left(m_0 c^2 \sqrt{1 - \frac{u^2}{c^2}} \right) \Big|_0^u = \frac{m_0 u^2}{\sqrt{1 - \frac{u^2}{c^2}}} + \left(m_0 c^2 \sqrt{1 - \frac{u^2}{c^2}} - m_0 c^2 \right) \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}$$

$$E_k = \frac{m_0 c^2}{\sqrt{1 - \frac{u^2}{c^2}}} - m_0 c^2 \rightarrow E_k = \gamma m_0 c^2 - m_0 c^2 = m_0 c^2 (\gamma - 1)$$

$$E_k + m_0 c^2 = \gamma m_0 c^2 \quad \boxed{m_0 c^2 = E}$$

Total energy Rest mass energy

Relativity tells us that even $v \ll c$, $E \neq 0$, mass is equivalent energy

This expression should reduce to the classical KE formula

$$E_k = \gamma m_0 c^2 - m_0 c^2 \quad \text{and} \quad \gamma = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}} \quad \text{take } -\frac{u^2}{c^2} = x$$

Use series expansion

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots, \quad \text{for } x \ll 1, \gamma = (1+x)^{-1/2}$$

$$E_k = \left(1 + \frac{1}{2} \frac{u^2}{c^2} \right) m_0 c^2 - m_0 c^2 \quad (\text{we neglect the terms higher than } O(x^2))$$

$$E_k = \frac{1}{2} m_0 u^2$$

We can also relate the total energy with momentum.

$$E = \gamma m_0 c^2, \quad p = \gamma m_0 u$$

$$E^2 - p^2 c^2 = \frac{m_0^2 c^4}{1 - \frac{u^2}{c^2}} - \frac{m_0^2 u^2 c^2}{1 - \frac{u^2}{c^2}} = (m_0 c^2)^2$$

is found after some algebra for photons with no mass $E = pc$

$$E^2 = (m_0 c^2)^2 + p^2 c^2$$

Total energy-momentum relation

E-P relation for massless particles.

In general, it's convenient to describe the rest mass of subatomic particles in terms of energy.

1 eV = Energy of an electron accelerated under 1V

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

$$E = m_e c^2 = (9.1 \times 10^{-31} \text{ kg}) (2.99 \times 10^8 \text{ m/s})^2 \text{ J} = (8.18 \times 10^{-14}) \text{ J}$$

$$E_e = (8.18 \times 10^{-14} \text{ J}) / (1.602 \times 10^{-19} \text{ J})$$

↳ electron mass energy

$$= 0.511 \times 10^6 \text{ eV}$$

$$E_e = 0.511 \text{ MeV}$$

Energy

Typically electrons travel in a TV tube with $u = 0.25c$ velocity. Find the total and the kinetic energy!

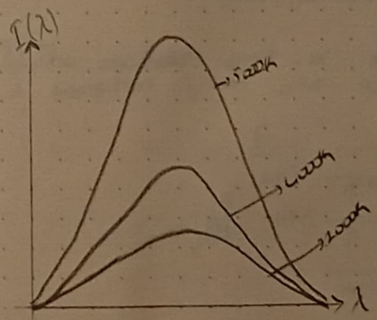
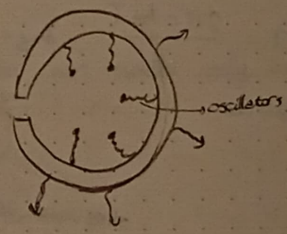
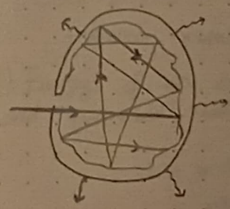
$$E = \gamma m_0 c^2 = \frac{m_0 c^2}{\sqrt{1 - \frac{u^2}{c^2}}} = \frac{0.511 \text{ MeV}}{\sqrt{1 - (0.25)^2}} = (1.03)(0.511) = 0.528 \text{ MeV}$$

$$E_k = E_T - m_0 c^2 = 0.528 - 0.511$$

$$E_k = 0.017 \text{ MeV}$$

Introduction to Quantum Physics

In the late 19th century there was an inconsistency between experimental and theoretical results about the thermal radiation from hot bodies



what we see from this plot

- 1) The total power of emitted radiation increases with temperature.

$$P = \sigma A T^4$$

↳ watts

↳ Stefan's law

↳ m^2 film

$$J = 5.67 \times 10^{-8} \text{ W/m}^2$$

In general, $P = \epsilon \sigma A T^4$ where $0 \leq \epsilon \leq 1$

For $\epsilon = 1$, this is a black body.

For $0 < \epsilon < 1$ this is a grey body.

2) The peak of the plot shifts to shorter wavelengths as the temperature increases.

$\lambda_{max} T = C$ Wien's displacement law

wavelength
of the peak
of the plot

corresponding
temperature

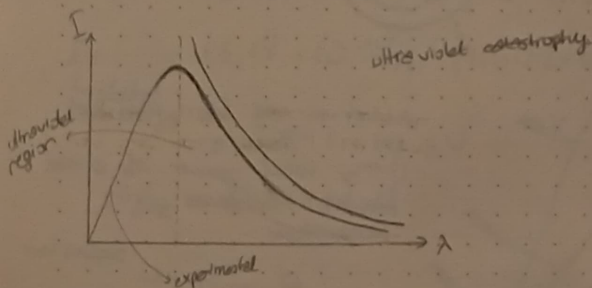
$$C_1 = 2.898 \times 10^{-3} \text{ mK}$$

What is the theory of $I(\lambda, T)$?

Based on the classical theory of radiation.

$$I(\lambda, T) = \frac{2\pi c^2 k_B T}{\lambda^4} \quad k_B = \text{Boltzmann constant}$$

Rayleigh-Jeans law



RJ formula can also be expressed in terms of energy density for frequencies

$$u(\nu) d\nu = \frac{8\pi h T}{c^3} \nu^2 d\nu \dots (v)$$

Total energy
per unit volume
in the frequency
interval between $\nu \rightarrow \nu + d\nu$

In 1900 Max Planck came up with a formula

$$u(\nu) d\nu = \frac{8\pi h}{c^3} \frac{\nu^3 d\nu}{e^{h\nu/kT} - 1}$$

Here, $h = 6.62 \times 10^{-34} \text{ Js}$

This formula perfectly
fit with experiment

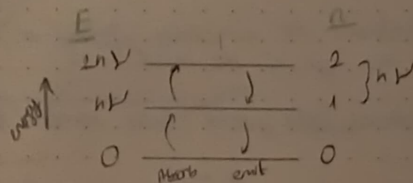
Assumptions Planck Made

1) Energy of an oscillator can have only certain discrete and "quantized" values E_n .

$$E_n = n h \nu$$

quantum number

2) The oscillators emit or absorb energy when making a transition from one quantum state to another.



Find the peak wavelength of the black-body radiation emitted by human body with the skin temperature

$$T = 35^\circ\text{C}$$

$$27^\circ\text{C} + 35^\circ\text{C} = 308\text{K}$$

$$\lambda_{\text{max}} T = C$$

$$\lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ mK}}{308 \text{ K}}$$

$$\lambda_{\text{max}} = 9.41 \times 10^{-6} \text{ m} \approx 9.4 \mu\text{m}$$

PHOTOELECTRIC EFFECT

It was known that metals were ejecting electrons when illuminated by light. Experiments show that

- 1) Constant light frequency (color) with different light intensity causes electron ejection with constant E_k with different number of electrons

classical prediction: higher light intensity should result with higher electron kinetic energy.

- 2) Constant light intensity with different light frequencies causes the number of ejected electrons remain constant while the E_k varies.

classical prediction: There should be no relationship between the light frequency and E_k

- 3) The electrons are sometimes ejected or not ejected depending on the type of the metal

Einstein introduced "light packets" later named "photons"

$$E = h\nu$$

photon energy

photon frequency

planck constant

$$\nu = \frac{c}{\lambda}, \quad E = \frac{hc}{\lambda}$$

wavelength dependent energy

$$10^{10} \text{ \AA} = 1 \text{ m}$$

Note

$$h = 6.62 \times 10^{-34} \text{ J s}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$hc = 19.86 \times 10^{-26} \text{ J m}$$

$$hf = 6.25 \times 10^{18} \text{ eV}$$

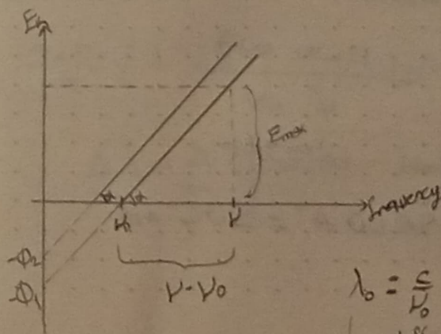
$$hc = (19.86 \times 10^{-26}) \times (6.25 \times 10^{18}) \times 10^{10}$$

$$hc \approx 12400 \text{ eV \AA}$$

The maximum kinetic energy of electrons

$$[E_k]_{\text{max}} = E_{\text{photon}} - \phi \rightarrow \text{metal work function}$$

$$E_k = h\nu - \phi$$



$$E_{\text{max}} = h(\nu - \nu_0)$$

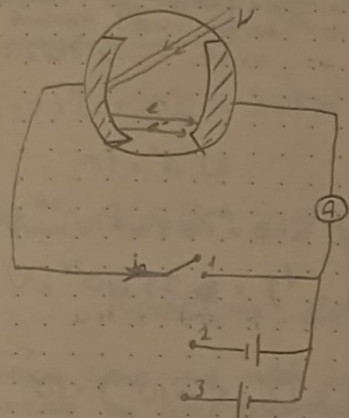
$$= h\nu - \frac{h\nu_0}{\phi}$$

$$E_{\text{max}} = h\nu - \phi$$

$$\lambda_0 = \frac{c}{\nu_0}$$

λ_0 cutoff wavelength

Model Example 2 (Photo cell)



①: 2: photoelectric current

$$h\nu = \phi + E_k$$

②: $h\nu + eV = \phi + E_k$

③: E field is against e's.

$$h\nu = eV + \phi + E_k$$

i.e. $eV + \phi > h\nu$ no current observed

A sodium surface is illuminated by light having 300nm wavelength. $\phi = 2.46$ eV

Find $[E_k]_{\max} = ?$ for electrons

Find the cutoff wavelength $\lambda_0 = ?$ for sodium

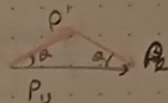
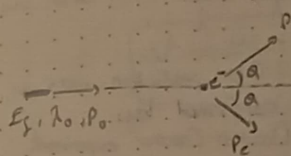
$$E_k = \frac{hc}{\lambda} - \phi = \frac{12400 \text{ eV}\cdot\text{\AA}}{3000 \text{\AA}} - 2.46 \text{ eV} = 1.67 \text{ eV}$$

$$\lambda_0 = \frac{c}{\nu_0} = \frac{hc}{h\nu_0} = \frac{12400 \text{ eV}\cdot\text{\AA}}{\phi \text{ eV}} \rightarrow 2.46$$

$$\lambda_0 = 5040 \text{\AA} = 504 \text{ nm}$$

Compton Effect

Arthur Compton stated that free electrons exhibit particle-like properties such as elastic scattering after interaction with light.



Using the conservation of momentum and energy

$$p_e^2 = p_0^2 + p'^2 + 2 \frac{E_0}{c} (p_0 - p') - 2 p_0 p' \cos \theta = p_0^2 + p'^2 - 2 p_0 p' \cos \theta$$

$$\frac{2 E_0}{c} (p_0 - p') = 2 p_0 p' - 2 p_0 p' \cos \theta$$

$$\frac{E_0}{c} (p_0 - p') = 2 p_0 p' (1 - \cos \theta)$$

$$\frac{E_0}{c} \frac{(p_0 - p')}{p_0 p'} = 1 - \cos \theta$$

$$\left(\frac{1}{p'} - \frac{1}{p_0} \right)$$

$$E = pc \rightarrow \frac{hc}{\lambda} = pc \rightarrow \frac{1}{\lambda} = \frac{p}{h}$$

$$\frac{m_e c}{h} (\lambda' - \lambda_0) = (1 - \cos \theta)$$

$$\Delta \lambda = \frac{h}{m_e c} (1 - \cos \theta)$$

This equation states that the wavelength of photon's depends only to the scattering angle θ

Matter Waves

Louis de Broglie stated that the wave and particle models of other matter or radiation complement each other

$$E = PC \quad P = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda} \quad \lambda = \frac{h}{p}$$

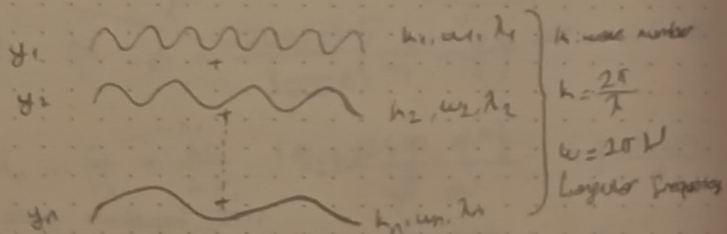
$$P = \frac{h}{\lambda} \Rightarrow \lambda = \frac{h}{p} \Rightarrow \lambda = \frac{h}{mv}$$

This prediction was confirmed by Davisson-Germer experiment

Modern Approach

(The wave-packet picture particle)

- An ideal particle is localized in space.
- An ideal wave is unlocalized in space and has a single frequency.
- However, a wave packet (which is a superposition of many waves) represents a particle which has also localization properties.



$$y_1 = A \cos(k_1 x - w_1 t)$$

$$y_2 = A \cos(k_2 x - w_2 t)$$

$$\cos a \cos b = 2 \cos \left[\frac{a+b}{2} \right] \cos \left[\frac{a-b}{2} \right]$$

$$y = 2A \cos \left[\frac{\Delta k}{2} x - \frac{\Delta w}{2} t \right] \cos \left(\frac{k_1 + k_2}{2} x - \frac{w_1 + w_2}{2} t \right)$$

$$\Delta k = k_1 - k_2 \quad \Delta w = w_1 - w_2$$

we note that $\lambda = vT \Rightarrow v = \lambda \nu = \frac{\lambda}{2\pi} \frac{2\pi\nu}{\lambda}$

$$v = \frac{w}{k} \text{ : phase speed}$$

Following this equation v = the ratio of time coefficient and the space coefficient of the wave packet.

$$v_g = \frac{\Delta w}{\Delta k} \rightarrow \frac{\partial w}{\partial k} = v_g$$

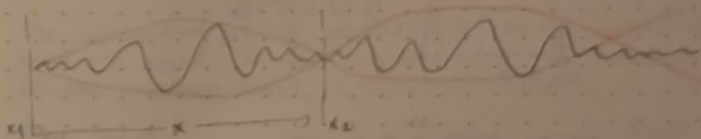
$$h = \frac{h}{2\pi}$$

$$E = h\nu = \frac{h 2\pi\nu}{2\pi}$$

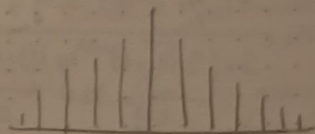
$$E = \frac{h w}{2\pi} = \hbar w$$

$$v_g = \frac{\partial (hw)}{\partial (hk)} = \frac{\partial E}{\partial p}$$

$$\frac{h}{2\pi} \cdot \frac{2\pi}{\lambda} = p = \hbar k$$



Probability of particle location



Heisenberg Uncertainty Principle

Werner Heisenberg stated that it is impossible to measure a particle's position and momentum simultaneously with infinite accuracy.

This result is independent from the measurement apparatus

$$\Delta x \cdot \Delta p_x \geq \frac{h}{2}$$

position uncertainty $\swarrow$ $\searrow$ momentum uncertainty

Uncertainty principle has a different form, $\Delta E \Delta t \geq \frac{h}{2}$

The energy of a quantum system cannot be measured precisely at finite time.

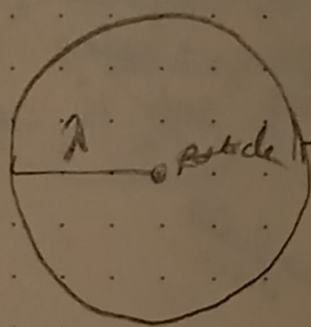
The speed of an e^- is measured to be 5×10^3 m/s to an accuracy of $3 \times 10^{-3} \%$. What is the minimum uncertainty determining the position of the particle?

$$\left. \begin{aligned} p_x \Delta x &= \frac{h}{2} \\ \Delta x &= \frac{h}{2m_e \Delta v} \end{aligned} \right\} \Delta v = v \cdot \Delta$$

$$\Delta x = \frac{1.05 \times 10^{-34} \text{ J s}}{2 \times \underbrace{(9.11 \times 10^{-31} \text{ kg})}_{m_e} \underbrace{(3 \times 10^{-4})}_{\Delta} \underbrace{(5 \times 10^3 \text{ m/s})}_v} = 3.86 \times 10^{-4} \text{ m}$$

We want to measure the position twice in succession but we don't want to measure the position with such a good determination that it gives a random kick to the velocity which is what we're trying to measure.

We wouldn't want to measure the velocity by beginning the experiment with a random kick which randomizes the velocity and sends it off in some direction very different than it was moving with to begin with. Both direction and velocity. So two measurements of position should be very gentle that don't change the momentum very much. That means they must be done with photons of very long wavelength. To give particle a good check, we do it with a very long wavelength photon.



Particle is in some region of size Δx
 $\Delta x \pm \lambda$

We changed the velocity by very small amount by using a very long wavelength photon.

Then the particle is at $x \pm \lambda + vt$

$$vt \pm \lambda \sim d$$

$$v \pm \frac{\lambda}{t} \sim \frac{d}{t}$$

→ if we take larger t , we can measure the velocity with great precision, however it's been at the cost of knowing where the particle is.

$|a\rangle$ ket vector

$\alpha |a\rangle = |b\rangle$ multiplying by alpha maps every vector into another
 $\alpha \in \mathbb{C}$

vector space over complex numbers is Hilbert Space

$$\psi(x) = \psi_A(x) + \psi_I(x)$$

ψ a complex function

The coin is a two-state system, the classical coin is described by a system with two states: Heads and Tails. And they form a set. That set is called the phase state.

$\{H, T\}$

→ confused coin

$$\alpha_H |H\rangle + \alpha_T |T\rangle = |CC\rangle$$

$$P_H = \alpha_H^* \alpha_H$$

$$P_H + P_T = 1$$

$$P_T = \alpha_T^* \alpha_T$$

$$\alpha_H^* \alpha_T + \alpha_T^* \alpha_T = 1 = \langle CC | CC \rangle$$